

and so on. These primitives allow programmers to control access to a shared resource. In addition, similar to processes, threads too suffer states of deadlock, or starvation, if not designed carefully. Debugging and analysing a threaded application can also be a little cumbersome.

How does Linux implement threads?

Linux supports the development and execution of multi-threaded applications. User-level threads in Linux follow the open POSIX (Portable Operating System Interface for uniX) standard, designated as IEEE 1003. The user-level library (on Ubuntu, *glibc.so*) has an implementation of the POSIX API for threads. Threads exist in two separate execution spaces in Linux—in user space and the kernel. User-space threads are created with the '*pthread*' library API (POSIX compliant). These user-space threads are mapped to kernel threads. In Linux, kernel threads are regarded as 'light-weight processes'. An LWP is the unit of a basic execution context. Unlike other UNIX variants, including HP-UX and SunOS, there is no special treatment for threads. A process or a thread in Linux is treated as a 'task', and shares the same structure representation (list of *struct task_struct*).

For a set of user threads created in a user process, there is a set of corresponding LWPs in the kernel. The following example illustrates this point:

```
#include <stdio.h>
#include <syscall.h>
#include <pthread.h>

int main()
{
    pthread_t tid = pthread_self();
    int sid = syscall(SYS_gettid);
    printf("LWP id is %d\n", sid);
    printf("POSIX thread id is %d\n", tid);
    return 0;
}
```

Running the '*ps*' command too,

lists processes and their LWP/ threads information:

```
kanaujia@ubuntu:~/Desktop$ ps -fl
UID      PID      PPID     LWP
C  NLPW  STIME  TTY          TIME CMD
kanaujia 17281    5191    17281  0   1
Jun11    pts/2    00:00:02 bash
kanaujia 22838    17281    22838  0   1
00:47    pts/2    00:00:00 ps -fl
kanaujia 17647    14111    17647  0   2
00:06    pts/0    00:00:00 vi clone.s
```

What is a Light-Weight Process?

An LWP is a process created to facilitate a user-space thread. Each user-thread has a 1x1 mapping to an LWP. The creation of LWPs is different from an ordinary process; for a user process 'P', its set of LWPs share the same group ID. Grouping them allows the kernel to enable resource sharing among them (resources include the address space, physical memory pages (VM), signal handlers and files). This further enables the kernel to avoid context switches among these processes. Extensive resource sharing is the reason these processes are called light-weight processes.

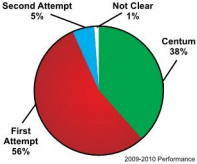
How does Linux create LWPs?

Linux handles LWPs via the non-standard *clone()* system call. It is similar to *fork()*, but more generic. Actually, *fork()* itself is a manifestation of *clone()*, which allows programmers to choose the resources to share between processes. The *clone()* call creates a process, but the child process shares its execution context with the parent, including the memory, file descriptors and signal handlers. The *pthread* library too uses *clone()* to implement threads. Refer to *"/nptl/sysdeps/pthread/createthread.c"* in the *glibc* version 2.11.2 sources.

Create your own LWP

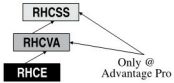
I will demonstrate a sample use of the *clone()* call. The program in *clone.c*


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